1st Midterm · Unit 4 · Linear systems and phase portraits · 5 points · 30 min · Standard exam level

P07 — Non-homogeneous planar system with a defective matrix: Jordan form, equilibrium shift, IVP and the improper node

Consider the system

$$x' = -x + y, \qquad y' = -x - 3y + 4.$$

(a) Write it as $\mathbf{x}' = A\mathbf{x} + \mathbf{f}$. Show that A has a repeated eigenvalue with only one independent eigenvector $\mathbf{v}$. Find a generalised eigenvector $\mathbf{w}$ with $(A - \lambda I)\mathbf{w} = \mathbf{v}$, build $P = [\mathbf{v} \mid \mathbf{w}]$, compute P^{-1} and check that $P^{-1}AP$ is the Jordan form.

(b) Find a particular solution by undetermined coefficients (a constant vector) and the general solution.

(c) Solve the initial value problem $x(0) = 3$, $y(0) = 0$ and check your answer.

(d) Classify the equilibrium and describe the phase portrait: invariant line, direction of tangency as $t \to \pm\infty$, sense of turning (test vector), nullclines, and the behaviour of the orbit of (c).

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknowns $x(t), y(t), t \in \mathbb{R}$. First-order, linear, constant coefficients, **non-homogeneous** with a constant forcing; solutions exist for all t and are unique given $\mathbf{x}(0)$.
- Structure of the general solution (superposition): $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$, with $\mathbf{x}_h$ the general solution of $\mathbf{x}' = A\mathbf{x}$.
- Since $\mathbf{f}$ is constant and 0 will not be an eigenvalue ($\det A \neq 0$), a constant $\mathbf{x}_p$ exists: it is the equilibrium, and $\mathbf{u} = \mathbf{x} - \mathbf{x}_p$ obeys the homogeneous system.

Phase 2. Canonical formulation

$$\mathbf{x}' = A\mathbf{x} + \mathbf{f}, \qquad A = \begin{pmatrix} -1 & 1 \\ -1 & -3 \end{pmatrix}, \qquad \mathbf{f} = \begin{pmatrix} 0 \\ 4 \end{pmatrix}.$$

Target frame: $\mathbf{u} = P\mathbf{z}$ with $P = [\mathbf{v} \mid \mathbf{w}]$, which turns $\mathbf{u}' = A\mathbf{u}$ into $\mathbf{z}' = J\mathbf{z}$, $J = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$.

Part (a) · 1.5 points

Repeated eigenvalue, generalised eigenvector and Jordan form

Step 1. Characteristic polynomial

$$\det(A-\lambda I) = (-1-\lambda)(-3-\lambda) - (1)(-1) = \lambda^2 + 4\lambda + 3 + 1 = \lambda^2 + 4\lambda + 4 = (\lambda+2)^2$$

so $\lambda = -2$ is a double root ($\operatorname{tr} A = -4 = 2\lambda$, $\det A = 4 = \lambda^2$).

Step 2. Eigenvectors

$N = A + 2I = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$. Row 1: $v_1 + v_2 = 0$; row 2: $-v_1 - v_2 = 0$, the same equation. The rank of N is 1, so the eigenspace is one-dimensional, spanned by $\mathbf{v} = (1,-1)^T$: A is not diagonalisable. Also $N^2 = \begin{pmatrix} 1-1 & 1-1 \\ -1+1 & -1+1 \end{pmatrix} = 0$, as the Cayley-Hamilton theorem predicts.

Step 3. Generalised eigenvector

$N\mathbf{w} = \mathbf{v}$: row 1 $w_1 + w_2 = 1$, row 2 $-w_1 - w_2 = -1$ (the same). Any solution works; take $w_2 = 0$, $w_1 = 1$: $\mathbf{w} = (1,0)^T$. It is not a multiple of $\mathbf{v}$, as required.

Step 4. Change of basis and Jordan form

$P = [\mathbf{v} \mid \mathbf{w}] = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}$, $\det P = 1\cdot 0 - 1\cdot(-1) = 1$, $P^{-1} = \frac{1}{1}\begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$.

$A\mathbf{v} = -2\mathbf{v} = (-2,2)^T$ and $A\mathbf{w} = (-1,-1)^T = \mathbf{v} - 2\mathbf{w}$, so $AP = \begin{pmatrix} -2 & -1 \\ 2 & -1 \end{pmatrix}$. Then

$$\begin{aligned} (P^{-1}AP)_{11} &= 0(-2) + (-1)(2) = -2, & (P^{-1}AP)_{12} &= 0(-1) + (-1)(-1) = 1, \\ (P^{-1}AP)_{21} &= 1(-2) + 1(2) = 0, & (P^{-1}AP)_{22} &= 1(-1) + 1(-1) = -2. \end{aligned}$$

$$P^{-1}AP = \begin{pmatrix} -2 & 1 \\ 0 & -2 \end{pmatrix} = J.$$

KEY RESULTS

- $p(\lambda) = \lambda^2 + 4\lambda + 4 = (\lambda + 2)^2$, $\lambda = -2$ (double); $N = A + 2I = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$, rank 1, $N^2 = 0$.
- $\mathbf{v} = (1, -1)^T$, $\mathbf{w} = (1, 0)^T$.
- $P = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}$, $\det P = 1$, $P^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$, $P^{-1}AP = \begin{pmatrix} -2 & 1 \\ 0 & -2 \end{pmatrix}$.

Part (b) · 1.5 points

Particular solution and general solution

Step 1. Undetermined constant vector

With $\mathbf{x}_p = (p, q)^T$ constant, $\mathbf{x}_p' = \mathbf{0}$ and the system gives $A\mathbf{x}_p = -\mathbf{f}$:

$$-p + q = 0, \qquad -p - 3q + 4 = 0.$$

The first gives $q = p$; substituting, $-4p + 4 = 0$, so $p = q = 1$. The solution is unique because $\det A = 4 \neq 0$. So $\mathbf{x}^* = (1, 1)^T$ is the only equilibrium.

Step 2. Homogeneous part in the Jordan frame

Let $\mathbf{u} = \mathbf{x} - \mathbf{x}^*$; then $\mathbf{u}' = \mathbf{x}' = A\mathbf{x} + \mathbf{f} = A\mathbf{u} + (A\mathbf{x}^* + \mathbf{f}) = A\mathbf{u}$. With $\mathbf{u} = P\mathbf{z}$ (P constant): $\mathbf{z}' = P^{-1}AP\,\mathbf{z} = J\mathbf{z}$, i.e.

$$z_1' = -2z_1 + z_2, \qquad z_2' = -2z_2.$$

The second equation gives $z_2 = c_2 e^{-2t}$. The first becomes $z_1' + 2z_1 = c_2 e^{-2t}$; multiply by the integrating factor e^{2t}: $(e^{2t} z_1)' = e^{2t} z_1' + 2e^{2t} z_1 = c_2$ (product rule). Integrating, $e^{2t} z_1 = c_1 + c_2 t$, so $z_1 = (c_1 + c_2 t) e^{-2t}$.

Step 3. Back to the original frame

$\mathbf{u} = P\mathbf{z} = z_1 \mathbf{v} + z_2 \mathbf{w} = c_1 e^{-2t}\mathbf{v} + c_2 e^{-2t}(t\mathbf{v} + \mathbf{w})$, the formula of Robinson Ch. 30. Adding $\mathbf{x}^*$:

$$\mathbf{x}(t) = \begin{pmatrix} 1 + (c_1 + c_2 + c_2 t) e^{-2t} \\ 1 - (c_1 + c_2 t) e^{-2t} \end{pmatrix}.$$

The fundamental matrix $\Phi = \left[e^{-2t}\mathbf{v} \mid e^{-2t}(t\mathbf{v} + \mathbf{w}) \right]$ has $\det \Phi = e^{-4t} \det[\mathbf{v} \mid t\mathbf{v} + \mathbf{w}] = e^{-4t} \det[\mathbf{v} \mid \mathbf{w}] = e^{-4t} \neq 0$, in agreement with Abel's formula for systems (**extension**) $W(t) = W(0) e^{(\operatorname{tr} A) t}$.

KEY RESULTS

- $\mathbf{x}_p = \mathbf{x}^* = (1, 1)^T$.
- $\mathbf{x}(t) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_1 e^{-2t} \begin{pmatrix} 1 \\ -1 \end{pmatrix} + c_2 e^{-2t} \left[t \begin{pmatrix} 1 \\ -1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right]$.
- In components: $x = 1 + (c_1 + c_2 + c_2 t) e^{-2t}$, $y = 1 - (c_1 + c_2 t) e^{-2t}$.

Part (c) · 1 point

Initial value problem x(0) = 3, y(0) = 0

Step 1. Constants

$x(0) = 1 + c_1 + c_2 = 3$ and $y(0) = 1 - c_1 = 0$. Hence $c_1 = 1$ and $c_2 = 2 - c_1 = 1$:

$$x(t) = 1 + (2 + t) e^{-2t}, \qquad y(t) = 1 - (1 + t) e^{-2t}.$$

Step 2. Check

By the product rule, $x' = e^{-2t} - 2(2 + t) e^{-2t} = -(3 + 2t) e^{-2t}$ and $y' = -e^{-2t} + 2(1 + t) e^{-2t} = (1 + 2t) e^{-2t}$. Right-hand sides:

$$-x + y = -1 - (2 + t) e^{-2t} + 1 - (1 + t) e^{-2t} = -(3 + 2t) e^{-2t},$$

$$-x - 3y + 4 = -1 - (2 + t) e^{-2t} - 3 + 3(1 + t) e^{-2t} + 4 = (1 + 2t) e^{-2t}.$$

Both equations hold, and $x(0) = 3$, $y(0) = 0$.

KEY RESULTS

- $c_1 = 1, c_2 = 1$.
- $x(t) = 1 + (2 + t) e^{-2t}$, $y(t) = 1 - (1 + t) e^{-2t}$.
- $x' = -(3 + 2t) e^{-2t}$, $y' = (1 + 2t) e^{-2t}$.

Part (d) · 1 point

Classification and phase portrait

Step 1. Type

A double negative eigenvalue with a one-dimensional eigenspace gives a stable improper node (Robinson Ch. 31): it lies on the parabola $(\mathrm{tr})^2 = 4\det$ of the trace-determinant plane, at the border between the nodes and the foci.

Step 2. Eigen-direction and tangency

The line $\mathbf{x}^* + s\mathbf{v}$, i.e. $x + y = 2$, is invariant ($c_2 = 0$: $\mathbf{u} = c_1 e^{-2t}\mathbf{v}$ moves straight to $(1,1)$). For $c_2 \neq 0$, $\mathbf{u} = e^{-2t}\big[(c_1 + c_2 t)\mathbf{v} + c_2\mathbf{w}\big]$ and the term $c_2 t\mathbf{v}$ dominates as $t \to \pm\infty$: the direction of $\mathbf{u}$ tends to $\pm\mathbf{v}$ at both ends. So every orbit enters $(1,1)$ tangent to the line $x + y = 2$, and far away (as $t \to -\infty$) it is again parallel to that line, on the other side: each orbit turns half a revolution.

Step 3. Sense of turning

With $\mathbf{u}' = A\mathbf{u}$: $u_1' = -u_1 + u_2$, $u_2' = -u_1 - 3u_2$, so

$$u_1 u_2' - u_2 u_1' = u_1(-u_1 - 3u_2) - u_2(-u_1 + u_2) = -u_1^2 - 2u_1u_2 - u_2^2 = -(u_1 + u_2)^2 \leq 0,$$

zero only on the invariant line. The polar angle around $(1,1)$ never increases: orbits turn **clockwise**. Test vector: at $(2,1)$, $\mathbf{u} = \mathbf{w} = (1,0)$, and $\mathbf{u}' = A\mathbf{w} = (-1,-1)^T$ points down, crossing the horizontal through $(1,1)$ downwards on its right: clockwise.

Step 4. Nullclines

$x' = 0$ on the line $y = x$ (vertical slopes there); $y' = 0$ on $x + 3y = 4$ (horizontal slopes). Both pass through $(1,1)$ and split the plane into four sectors where (x', y') keeps its signs; e.g. at $(3,0)$: $x' = -3 < 0$, $y' = 1 > 0$.

Step 5. The orbit of (c)

$u_1 + u_2 = (2+t)e^{-2t} - (1+t)e^{-2t} = e^{-2t} > 0$: the orbit never meets the invariant line and stays in the half-plane $x + y > 2$ (uniqueness forbids crossing an invariant line). For $t \geq 0$, $x' = -(3+2t)e^{-2t} < 0$ and $y' = (1+2t)e^{-2t} > 0$: it moves monotonically from $(3,0)$ to $(1,1)$, with $u_2/u_1 = -(1+t)/(2+t) \to -1$, i.e. arriving from the lower right tangent to $x + y = 2$. Backwards in time it crosses $y = 1$ at $t = -1$ and $x = 1$ at $t = -2$, and escapes to infinity parallel to $(-1,1)$.

KEY RESULTS

- $(1,1)$ is a **stable improper node**: $\operatorname{tr} A = -4 < 0$, $\det A = 4 > 0$, $(\operatorname{tr} A)^2 - 4\det A = 0$, one eigen-direction.
- Invariant line $x + y = 2$ (through $(1,1)$ along $\mathbf{v} = (1,-1)$); all other orbits are tangent to it at $(1,1)$ and parallel to it as $t \to -\infty$.
- $u_1 u_2' - u_2 u_1' = -(u_1 + u_2)^2 \le 0$: orbits turn **clockwise**; test vector $A\mathbf{w} = (-1,-1)^T$ at $(2,1)$.
- Nullclines: $x' = 0$ on $y = x$; $y' = 0$ on $x + 3y = 4$. The orbit of (c) stays in $x + y > 2$; for $t \ge 0$ x decreases to 1 and y increases to 1.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** $t \to +\infty$: $\mathbf{x} \to (1,1)$ like te^{-2t}, slower than a pure e^{-2t} because of the secular factor t. $t \to -\infty$: $|\mathbf{x}| \to \infty$. If $\mathbf{f} = \mathbf{0}$ the portrait is the same, centred at the origin.
- **Checks:** $P^{-1}AP = J$ with $\det P = 1$; $A\mathbf{x}^* + \mathbf{f} = \mathbf{0}$; the IVP solution satisfies both equations and the initial data; $\det \Phi = e^{-4t}$ matches Abel's formula for systems (extension).
- **Common mistakes in the exam:** (1) writing $\mathbf{x}_h = c_1 e^{-2t}\mathbf{v} + c_2 te^{-2t}\mathbf{v}$, which forgets $\mathbf{w}$ and is not a fundamental set; (2) solving $(A - \lambda I)\mathbf{w} = \mathbf{v}$ with the wrong sign of λ (using $A - 2I$); (3) trying a particular solution $\mathbf{c}\,t$ or $\mathbf{c}\,e^{-2t}$ for a constant forcing: a constant vector suffices since $\det A \neq 0$; (4) drawing a star node (all straight lines), or claiming orbits cross the eigenline.